1. Clinical Question & Core Hypothesis

Clinical Question

PROS1 is expressed in tumor tissue and associated with Ki-67, but it is unclear whether PROS1 functionally drives proliferation and whether an ultrasound-visible tumor phenotype captures this biology and predicts clinical progression. 2) Tier 1 path used: the single provided path linking PROS1 expression to Ki-67 (MKI67) — I built on this because it gives a direct molecular–proliferation axis. 3) Supporting evidence: the path notes ultrasound should be included in bladder evaluation and that ultrasound exposure does not alter gene expression, so imaging can be used as a non-confounding phenotyping step to identify tumors for transcriptional/proliferative correlation. 4) Endpoint choice: I selected progression-free interval (PFI) because proliferation (MKI67) most directly drives early progression/recurrence dynamics in solid tumors, making PFI the most mechanistically relevant clinical outcome. 5) Deliberately excluded: I did not invoke downstream receptors/pathways (e.g., TAM receptors) or additional imaging features (e.g., TIL/TLS/TSR) to keep the hypothesis sharply focused on whether PROS1 causally promotes proliferation as captured by ultrasound-visible phenotype; no other genes are included in the core hypothesis to avoid overfitting the mechanistic claim?

Working Hypothesis

PROS1 upregulation in bladder tumor cells activates a pro-proliferative transcriptional program that increases MKI67 expression and enhances cell-cycle progression in ultrasound-visible tumors; consequently, ultrasound-visible, PROS1-high bladder tumors will have shorter progression-free intervals than ultrasound-visible, PROS1-low tumors, independent of treatment and imaging timing, provided ultrasound does not alter PROS1 or MKI67 expression.

Element	Value
Imaging feature	Ultrasound
Anchor gene(s)	PROS1, MKI67
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 is expressed in tumor tissue and associated with Ki-67, but it is unclear whether PROS1 functionally drives proliferation and whether an ultrasound-visible tumor phenotype captures this biology and predicts clinical progression.

MKI67: MKI67) most directly drives early progression/recurrence dynamics in solid tumors, making PFI the most mechanistically relevant clinical outcome.

3. Clinically Testable Predictions

Prediction 1: In vitro: manipulate PROS1 (overexpression/knockdown) in bladder cancer cell lines; measure MKI67 levels, proliferation rates (EdU incorporation, Ki-67 staining), and cell-cycle profiles;

Prediction 2: Transcriptomics: perform RNA-seq to define PROS1-associated pro-proliferative programs and validate MKI67 correlation;

Prediction 3: In vivo: orthotopic bladder cancer models with altered PROS1; monitor growth with ultrasound, categorize lesions as ultrasound-visible, and compare time-to-progression metrics;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions